

HODGE DECOMPOSITION AND THE DEGENERATION OF THE FRÖLICHER SPECTRAL SEQUENCE

YANGDA BEI

ABSTRACT. In this expository paper, we discuss the Hodge decomposition theorem on the cohomology of compact Kähler manifolds. In particular, we compare two different forms of the theorem: one which arises from the study of harmonic forms, and the other as the degeneration of the Frölicher spectral sequence.

1. INTRODUCTION

In complex geometry, Hodge theory provides a variety of tools for studying the geometry and topology of complex manifolds. An important result of Hodge theory is the following decomposition theorem:

Theorem 1.1 (Hodge decomposition theorem). *If X is a compact Kähler manifold, there is a decomposition of the cohomology of X with complex coefficients as a direct sum of vector spaces:*

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X).$$

We can reformulate Theorem 1.1 as a theorem of the degeneracy of the Frölicher spectral sequence.

Theorem 1.2. *If X is a compact Kähler manifold, the spectral sequence induced by the naïve filtration on the holomorphic de Rham complex*

$$E_i^{p,q} \Rightarrow \mathbb{H}^{p,q}(X, \Omega_{hol}^\bullet) = H^{p+q}(X, \mathbb{C})$$

degenerates on the first page.

This formulation of Theorem 1.1 is surprisingly actually a statement of algebraic geometry (in the case of projective manifolds) via Serre's GAGA principle [Ser56] (see [Bei24] for an exposition of the proof in the algebraic case by the author). Now, the degeneracy of the Frölicher spectral sequence does not fully imply the Hodge decomposition theorem. For non-Kähler manifolds, non-degenerating differentials in higher steps show how much Dolbeault cohomology differs from de Rham cohomology, and in some sense, how far a complex manifold is from being Kähler. However, for compact surfaces, Kähler or not, the Frölicher spectral sequence degenerates at E_1 [BHPVdV15, Theorem 2.8], and from that, one can understand to what extent Hodge decomposition is valid.

The aim of this paper is to present the construction and implications of Theorem 1.1. In Section 2, we begin by presenting the basic notions of complex geometry and introducing the Dolbeault cohomology groups of a complex manifold. We then reinterpret de Rham and Dolbeault cohomology in terms of sheaf cohomology, and introduce the concept of hypercohomology in Section 3. Section 4 is a brief summary of Kähler manifolds and some important results, eventually leading to the statement of the Hodge decomposition theorem. We conclude the paper in Section 5 with a discussion the degeneration of the Frölicher spectral sequence, comparing the induced Hodge filtration from the spectral sequence and the filtration which arises from the Hodge structures of compact Kähler manifolds.

2. BASIC CONSTRUCTIONS IN COMPLEX GEOMETRY

2.1. Complex manifolds and vector bundles.

Definition 2.1. A $2n$ -(real)-dimensional manifold X is a n -dimensional complex manifold if it admits an open cover $\{U_i\}$ and charts $\{(U_i, \phi_i)\}$ where the maps $\phi_i: U_i \rightarrow \mathbb{C}^n$ are smooth (when $\mathbb{C}^n$ is identified with $\mathbb{R}^{2n}$), in such a way that the transition functions $\phi_j \circ \phi_i^{-1}: \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j)$ are holomorphic for all i, j .

Definition 2.2. Given a complex manifold X , the *complexified tangent space* to X at p is defined to be the complexification of the (real) tangent space at p , that is

$$T_{\mathbb{C},p}(X) := T_{\mathbb{R},p}(X) \otimes \mathbb{C}.$$

These form the fibres of a complex vector bundle $T_{\mathbb{C}}(X)$ with (complex) rank $2n$, which we call the *complexified tangent bundle*. Similarly, we may define the *cotangent vector space* to be the complexification of the (real) cotangent space at p , that is

$$T_{\mathbb{C},p}^*(X) := T_{\mathbb{R},p}^*(X) \otimes \mathbb{C}.$$

These form the fibres of a rank $2n$ complex vector bundle $T_{\mathbb{C}}^*(X)$, which we call the *complexified cotangent bundle*.

Noticing that $T_{\mathbb{C},p}^*(X) \cong \text{Hom}(T_{\mathbb{C},p}(X), \mathbb{C})$, the complexified cotangent space is dual to the complexified tangent space as one would expect. Fix a point p , and let $\phi = (z_1, \dots, z_n): U \rightarrow \mathbb{C}^n$ be holomorphic coordinates near p . As a real vector space, $\mathbb{C}^n$ is canonically identified with $\mathbb{R}^{2n}$ under the map $(a_1 + ib_1, \dots, a_n + ib_n) \mapsto (a_1, \dots, a_n, b_1, \dots, b_n)$. By composing ϕ with this map, we obtain C^∞ coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ with $z_j = x_j + iy_j$. Thus, $T_{\mathbb{R},p}(X) = \text{span}_{\mathbb{R}}\langle \partial_{x_j}, \partial_{y_j} \rangle$. Letting $\partial_{z_j} := \frac{1}{2}(\partial_{x_j} - i\partial_{y_j})$ and $\partial_{\bar{z}_j} := \frac{1}{2}(\partial_{x_j} + i\partial_{y_j})$, we see that $T_{\mathbb{C},p}(X) = \text{span}_{\mathbb{C}}\langle \partial_{z_j}, \partial_{\bar{z}_j} \rangle$. An important remark is that if $f: X \rightarrow \mathbb{C}$ is a smooth function, then f being holomorphic (resp. antiholomorphic) is equivalent to $\partial_{\bar{z}_j}f = 0$ (resp. $\partial_{z_j}f = 0$) for all $1 \leq j \leq n$, which follows easily from the Cauchy-Riemann equations.

In particular, the complexified tangent space $T_{\mathbb{C},p}(X)$ splits as the direct sum of the subspaces

$$T_p^{1,0}(X) := \text{span}_{\mathbb{C}}\langle \partial_{z_j} \rangle \quad \text{and} \quad T_p^{0,1}(X) := \text{span}_{\mathbb{C}}\langle \partial_{\bar{z}_j} \rangle.$$

This fibre-wise splitting induces a splitting on vector bundles $T_{\mathbb{C}}(X) = T^{1,0}(X) \oplus T^{0,1}(X)$. We call $T^{1,0}(X)$ the *holomorphic tangent bundle* and $T^{0,1}(X)$ the *antiholomorphic tangent bundle*.

Letting dz_j and $d\bar{z}_j$ be the duals of ∂_{z_j} and $\partial_{\bar{z}_j}$ respectively, we obtain an induced dual decomposition of the complexified cotangent bundle $T_{\mathbb{C}}^*(X) = \Omega^{1,0}(X) \oplus \Omega^{0,1}(X)$, where $\Omega_p^{1,0}(X) = \text{span}_{\mathbb{C}}\langle dz_j \rangle$ and $\Omega_p^{0,1}(X) = \text{span}_{\mathbb{C}}\langle d\bar{z}_j \rangle$ on the fibres. Denoting $\Omega^{p,q}(X) := \bigwedge^p \Omega^{1,0}(X) \otimes \bigwedge^q \Omega^{0,1}(X)$, the complexified k -forms $\Omega_{\mathbb{C}}^k(X)$ (where on fibres, $\Omega_{\mathbb{C},p}^k(X) := \Omega_{\mathbb{R},p}^k(X) \otimes \mathbb{C}$) has the induced decomposition

$$\Omega_{\mathbb{C}}^k(X) = \bigoplus_{p+q=k} \Omega^{p,q}(X).$$

2.2. Dolbeault and de Rham cohomology. Let $d: \Omega_{\mathbb{R}}^k(X) \rightarrow \Omega_{\mathbb{R}}^{k+1}(X)$ denote the de Rham differential. We can extend d to a $\mathbb{C}$ -linear differential on the complexified k -forms $d: \Omega_{\mathbb{C}}^k(X) \rightarrow \Omega_{\mathbb{C}}^{k+1}(X)$. In local coordinates $(z_1, \dots, z_n)$, for a function f (equivalently a form of type $(0,0)$), we have that

$$d(f dz_I \wedge d\bar{z}_J) = \sum_{i=1}^n \partial_{z_i}(f) dz_i \wedge dz_I \wedge d\bar{z}_J + \sum_{j=1}^n \partial_{\bar{z}_j}(f) d\bar{z}_j \wedge dz_I \wedge d\bar{z}_J,$$

where I and J are the sets of ordered indices $i \leq i_1 < \dots < i_p \leq n$ and $1 \leq j_1 < \dots < j_q \leq n$, and $dz_I := dz_{i_1} \wedge \dots \wedge dz_{i_p}$ and $d\bar{z}_J := d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}$. Thus,

$$d(\Omega^{p+q}(X)) \subseteq \Omega^{p+1,q}(X) \oplus \Omega^{p,q+1}(X),$$

so we may uniquely decompose d as $d = \partial + \bar{\partial}$, where $\partial(\Omega^{p,q}(X)) \subseteq \Omega^{p+1,q}(X)$ and $\bar{\partial}(\Omega^{p,q}(X)) \subseteq \Omega^{p,q+1}(X)$. We now define the complex de Rham cohomology groups.

Definition 2.3. The k -th de Rham cohomology group of a differentiable manifold X , and respectively a complex manifold, are defined to be

$$H_{DR}^k(X; \mathbb{R}) := \frac{\ker\{d: \Omega_{\mathbb{R}}^k(X) \rightarrow \Omega_{\mathbb{R}}^{k+1}(X)\}}{\text{im}\{d: \Omega_{\mathbb{R}}^{k-1}(X) \rightarrow \Omega_{\mathbb{R}}^k(X)\}} \quad \text{and} \quad H_{DR}^k(X; \mathbb{C}) := \frac{\ker\{d: \Omega_{\mathbb{C}}^k(X) \rightarrow \Omega_{\mathbb{C}}^{k+1}(X)\}}{\text{im}\{d: \Omega_{\mathbb{C}}^{k-1}(X) \rightarrow \Omega_{\mathbb{C}}^k(X)\}}.$$

Because the functor $- \otimes_{\mathbb{R}} \mathbb{C}$ is exact as $\mathbb{C}$ is a free $\mathbb{R}$ -module, hence flat, it commutes with kernels and cokernels. In particular, this means that it commutes with taking the cohomology of a complex, so we get the following lemma:

Lemma 2.4. The real and complex de Rham cohomology groups satisfy

$$H_{DR}^*(X; \mathbb{C}) = H_{DR}^*(X; \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}.$$

From the decomposition $d = \partial + \bar{\partial}$, we see that $0 = d^2 = \partial^2 + \partial \circ \bar{\partial} + \bar{\partial} \circ \partial + \bar{\partial}^2$. It follows that $\partial^2 = \partial \circ \bar{\partial} = \bar{\partial}^2 = 0$ from the direct sum decomposition of the image $\Omega_{\mathbb{C}}^k(X) = \bigoplus_{p+q=k} \Omega^{p,q}(X)$. For fixed p , $\Omega^{p,\bullet}(X)$ becomes a complex with differential $\bar{\partial}$.

Definition 2.5. The Dolbeault cohomology groups of a complex manifold X are

$$H^{p,q}(X) := \frac{\ker\{\bar{\partial}: \Omega^{p,q}(X) \rightarrow \Omega^{p,q+1}(X)\}}{\text{im}\{\bar{\partial}: \Omega^{p,q-1}(X) \rightarrow \Omega^{p,q}(X)\}}.$$

We define the Hodge numbers to be

$$h^{p,q}(X) := \dim_{\mathbb{C}} H^{p,q}(X).$$

Remark 2.6. One can also define a complex for ∂ , where all proofs about differential forms go through complex conjugation effortlessly, however $\bar{\partial}$ is used widely as it frames everything in terms of holomorphic functions.

3. SHEAF COHOMOLOGY AND HYPERCOHOMOLOGY

In this section, we generalise the constructions of de Rham and Dolbeault cohomology by interpreting them in the language of sheaves. One might try to provide a definition of sheaf cohomology by mimicking the construction of de Rham or Dolbeault cohomology. Indeed, we shall see that every sheaf has a canonical resolution, which allows us to obtain a well-defined sheaf cohomology theory. However, standard sheaf cohomology may not capture all the necessary information in some settings, such as the cohomology of the sheaf holomorphic forms for a complex manifold (which is a great deal of importance to algebraic geometers). We will see that hypercohomology is a natural generalisation of sheaf cohomology, allowing us to address these gaps.

We first recall the basic notions of sheaves. For a full treatment, see [Voi02, Section 4.1.1]. A *presheaf* $\mathcal{F}$ over a topological space X assigns an abelian group $\mathcal{F}(U)$ to each open set $U \subseteq X$, requiring $\mathcal{F}(\emptyset) = 0$, which is compatible with restrictions. A *morphism of presheaves* is a map between presheaves that commutes with restrictions. We call elements of $\mathcal{F}(U)$ *sections*. An important definition that allows us to examine local properties around a point is the following:

Definition 3.1. Let $\mathfrak{U}_x$ be the open sets containing a point $x \in X$. The *stalk* $\mathcal{F}_x$ of a presheaf $\mathcal{F}$ over X at x is the direct limit

$$\mathcal{F}_x := \varinjlim_{U \in \mathfrak{U}_x} \mathcal{F}(U)$$

taken over the set of open sets containing x , ordered by inclusion, where the morphisms $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ for $V \subset U$ are the restriction morphisms. We call an element of $\mathcal{F}_x$ a *germ of sections of $\mathcal{F}$ at x* .

Finally, a presheaf of abelian groups $\mathcal{F}$ is a *sheaf* if for every open set $U \subseteq X$, and for every open covering $\{U_i\}_{i \in I}$ of U , the natural map $\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i)$ induces an isomorphism of $\mathcal{F}(U)$ onto

$$\{(s_{U_i})_{i \in I} : s_{U_i}|_{U_i \cap U_j} = s_{U_j}|_{U_i \cap U_j}, \text{ for all } i, j \in I\}.$$

3.1. Sheaf cohomology. We now define sheaf cohomology of a sheaf $\mathcal{F}$ over a topological space X . Fix an open subset U of X . For every sheaf $\mathcal{F}$ on X , we can associate the abelian group $\Gamma(U, \mathcal{F}) := \mathcal{F}(U)$ and to every sheaf morphism $\phi: \mathcal{F} \rightarrow \mathcal{G}$ the group homomorphism $\phi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$. This makes $\Gamma(U, -)$ into a functor from the category of sheaves to abelian groups, henceforth called the *section functor*. Aptly, we call $\Gamma(X, -)$ the *global sections functor*.

Lemma 3.2. *The global sections functor is left-exact.*

We can now rephrase de Rham and Dolbeault cohomology via sheaves.

Example 3.3. Let X be an n -dimensional smooth manifold. The constant sheaf $\mathbb{R}$ is the sheaf of locally constant functions with values in $\mathbb{R}$, which is naturally included in the sheaf of C^∞ functions. Let $\mathcal{A}^k$ be the sheaf of C^∞ k -forms on X . The exterior derivative $d: \mathcal{A}^k(U) \rightarrow \mathcal{A}^{k+1}(U)$, defined for every open set U in X , induces a morphism of sheaves $d: \mathcal{A}^k \rightarrow \mathcal{A}^{k+1}$. Now, exactness of a sequence of sheaves on X is equivalent to exactness of the sequence of stalks at every point $x \in X$ (see [Voi02, Section 4.1.2]). We therefore see that the sequence of sheaves

$$0 \longrightarrow \mathbb{R} \hookrightarrow \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \xrightarrow{d} \dots \xrightarrow{d} \mathcal{A}^n \longrightarrow 0$$

is exact by Poincaré's lemma, so $\mathcal{A}^\bullet$ forms a resolution for $\mathbb{R}$. Applying the global sections functor to this sequence yields

$$0 \longrightarrow \Gamma(X, \mathbb{R}) \longrightarrow \mathcal{A}^0(X) \xrightarrow{d} \mathcal{A}^1(X) \xrightarrow{d} \dots \xrightarrow{d} \mathcal{A}^n(X) \longrightarrow 0.$$

Omitting the initial term $\Gamma(X, \mathbb{R})$ and taking the cohomology of the resulting complex yields the *de Rham cohomology* of X .

Example 3.4. If X is a complex manifold of complex dimension n , the analogue of the Poincaré lemma is the $\bar{\partial}$ -Poincaré lemma [Voi02, Proposition 2.31]. It follows that for each fixed integer $p \geq 0$, the sheaves $\mathcal{A}^{p,q}$ of C^∞ (p, q) -forms on X gives rise to a resolution of the sheaf Ω_{hol}^p of holomorphic p -forms (sections of the vector bundle $\bigwedge^p \Omega^{1,0}(X)$), resulting in the exact sequence of sheaves

$$0 \longrightarrow \Omega_{hol}^p \hookrightarrow \mathcal{A}^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1} \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \mathcal{A}^{p,n} \longrightarrow 0.$$

Again omitting the initial term $\Gamma(X, \Omega_{hol}^p)$ after taking the global sections functor, the cohomology of the resulting complex yields the *Dolbeault cohomology* of X . From now on, we will use $\Omega_{\mathbb{R}}^k(X)$ and $\mathcal{A}^k(X)$ (resp. $\Omega^{p,q}(X)$ and $\mathcal{A}^{p,q}(X)$) interchangeably.

We see from the previous two examples that to define the cohomology of a sheaf $\mathcal{F}$, we might wish to find a resolution for $\mathcal{F}$, apply the global sections functor, and take the cohomology of the resulting complex. Indeed, there is a functorial and injective resolution for any sheaf. If $\mathcal{F}$ is a sheaf on a topological space X ,

let $\mathcal{C}^0\mathcal{F}$ be the sheaf defined by $\mathcal{C}^0\mathcal{F}(U) := \bigoplus_{x \in U} \mathcal{F}_x$. If $\mathcal{Q}^1$ is the quotient sheaf of $\mathcal{C}^0\mathcal{F}$ of $\mathcal{F}$, then we have a short exact sequence of sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0\mathcal{F} \longrightarrow \mathcal{Q}^1 \longrightarrow 0.$$

Then let $\mathcal{C}^1\mathcal{F} := \mathcal{C}^0\mathcal{Q}^1$ and $\mathcal{Q}^2$ be the quotient of $\mathcal{C}^0\mathcal{Q}^1$ by $\mathcal{Q}^1$, which yields the short exact sequence

$$0 \longrightarrow \mathcal{Q}^1 \longrightarrow \mathcal{C}^1\mathcal{F} \longrightarrow \mathcal{Q}^2 \longrightarrow 0.$$

We can join the two short exact sequences into a four-term exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{C}^0\mathcal{F} & \dashrightarrow & \mathcal{C}^1\mathcal{F} \longrightarrow \mathcal{Q}^2 \longrightarrow 0. \\ & & & & \searrow & & \nearrow \\ & & & & & \mathcal{Q}^1 & \end{array}$$

Repeating the construction on $\mathcal{Q}^2$, and so on, we obtain the long exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0\mathcal{F} \xrightarrow{\delta} \mathcal{C}^1\mathcal{F} \xrightarrow{\delta} \mathcal{C}^2\mathcal{F} \xrightarrow{\delta} \dots$$

of sheaves. This yields the *Godement canonical resolution* of $\mathcal{F}$, and we call $\mathcal{C}^k(-)$ the *k-th Godement functor*. We can now state the definition of sheaf cohomology.

Definition 3.5. The *sheaf cohomology groups* $H^k(X, \mathcal{F})$ of a sheaf of abelian groups $\mathcal{F}$ on a topological space is

$$H^k(X, \mathcal{F}) = \frac{\ker \delta: \mathcal{C}^k\mathcal{F}(X) \rightarrow \mathcal{C}^{k+1}\mathcal{F}(X)}{\operatorname{im} \delta: \mathcal{C}^{k-1}\mathcal{F}(X) \rightarrow \mathcal{C}^k\mathcal{F}(X)}.$$

Remark 3.6. Experts of homological algebra may know that we can define the *i*-th sheaf cohomology group of $\mathcal{F}$ as

$$H^i(X, \mathcal{F}) := R^i\Gamma(X, \mathcal{F}),$$

where $R^i\Gamma$ is the *i*-th (right-)derived functor of Γ . This is well-defined as the category of sheaves has enough injective objects. In defining sheaf cohomology this way, we implicitly construct $R^i\Gamma$ using a choice of injective resolution in the category of sheaves of abelian groups. Although different choices of injective resolutions give a canonical isomorphism between the objects $R^i\Gamma(\mathcal{F})$ computed, they are unwieldy to manipulate practically. The Godement resolution is “canonical” in the sense that the construction of this resolution works for any sheaf.

Remark 3.7. If $\mathcal{U}$ is a good cover on a manifold X , we have that $\check{H}^i(\mathcal{U}, \mathbb{R}) \cong H^i(M, \mathbb{R})$, where $\check{H}^i(\mathcal{U}, \mathbb{R})$ is the Čech cohomology of the good cover $\mathcal{U}$ with values in $\mathbb{R}$. We can instead generalise this statement to *paracompact spaces*. These are topological spaces in which every open cover has an open refinement that is locally finite. Then for any open covering $\mathcal{U}$ of a paracompact space X equipped with a sheaf $\mathcal{F}$, there is an isomorphism $\check{H}^i(\mathcal{U}, \mathcal{F}) \cong H^i(X, \mathcal{F})$ [EZT13, Theorem 2.7.4].

3.2. Hypercohomology. The hypercohomology of a complex of sheaves $\mathcal{F}^\bullet$ generalises the cohomology of a single sheaf. We shall see that it is hypercohomology which allows us to describe the relationship between the Dolbeault and de Rham cohomology of a complex manifold.

As with normal sheaf cohomology, we can compute the hypercohomology of a complex of sheaves by taking the Godement canonical resolution, this time for every sheaf $\mathcal{F}^\bullet$. This yields the double complex

$$K = \bigoplus_{p,q} K^{p,q} = \bigoplus_{p,q} \Gamma(X, \mathcal{C}^p\mathcal{F}^q)$$

with the horizontal differential $\delta: K^{p,q} \rightarrow K^{p+1,q}$ induced from the Godement resolution and the vertical differential $d: K^{p,q} \rightarrow K^{p,q+1}$ induced from the complex $\mathcal{F}^\bullet$. The fact that these two differentials commute

follows since the differential $d: \mathcal{F}^q \rightarrow \mathcal{F}^{q+1}$ induces a morphism of complexes $\mathcal{C}^\bullet \mathcal{F}^1 \rightarrow \mathcal{C}^\bullet \mathcal{F}^{q+1}$ and that the Godement functor $\mathcal{C}^\bullet(-)$ from sheaves to sheaves is exact. We can define hypercohomology as follows:

Definition 3.8. The *hypercohomology* $\mathbb{H}^i(X, \mathcal{F}^\bullet)$ of a complex of sheaves $\mathcal{F}^\bullet$ over a topological space X is the cohomology of the double complex K , that is the single cohomology $H_D^i(K^\bullet)$ of the complex

$$K^\bullet = \bigoplus_i K^i = \bigoplus_i \bigoplus_{p+q=i} K^{p,q}$$

with differential $D = \delta + (-1)^p d$ as defined above.

If $\mathcal{F}^\bullet$ is the complex with a single sheaf $\mathcal{F}_0$ in degree 0 given by

$$0 \longrightarrow \mathcal{F}^0 \longrightarrow 0 \longrightarrow 0 \longrightarrow \dots,$$

then $\bigoplus K^{p,q} = \bigoplus \Gamma(X, \mathcal{C}^p \mathcal{F}^q)$ has non-zero entries in only the zeroth row. This is simply the complex of sections of the Godement resolution of $\mathcal{F}^0$, so in this case, $\mathbb{H}^i(X, \mathcal{F}^\bullet) = H^i(X, \mathcal{F}^0)$. It is in this sense that hypercohomology is a generalisation of sheaf cohomology.

Remark 3.9. We can similarly define the hypercohomology of X with coefficients in $\mathcal{F}^\bullet$ as the (right-)derived functor of the global sections functor

$$\mathbb{H}^i(X, \mathcal{F}^\bullet) := \mathbb{R}^i \Gamma(X, \mathcal{I}^\bullet),$$

where $\mathcal{I}^\bullet$ is an injective resolution of $\mathcal{F}^\bullet$. Again, the double complexes of sheaves with different injective resolutions are quasi-isomorphic.

Example 3.10. Just as sheaf cohomology can be calculated from the Čech complex, we can also compute hypercohomology using a similar method. Let $(\mathcal{F}^\bullet, d_{\mathcal{F}})$ be a complex of sheaves on a topological space X and $\mathfrak{U}$ an open cover of X . The *Čech-sheaf double complex* is the complex

$$K = \bigoplus K^{p,q} = \bigoplus \check{C}^p(\mathfrak{U}, \mathcal{F}^q).$$

The *Čech cohomology* $\check{H}^*(\mathfrak{U}, \mathcal{F}^\bullet)$ of $\mathcal{F}^\bullet$ is defined to be the cohomology of the single complex

$$K^\bullet = \bigoplus_i K^i = \bigoplus_i \bigoplus_{p+q=k} \check{C}^p(\mathfrak{U}, \mathcal{F}^q)$$

with differential $d_K = \delta + (-1)^p d_{\mathcal{F}}$. If $\check{H}(\mathfrak{U}, \mathcal{F}^q) = 0$ for all q and $k > 0$, then there is an isomorphism $\check{H}^i(\mathfrak{U}, \mathcal{F}^\bullet) \cong \mathbb{H}^i(X, \mathcal{F}^\bullet)$ (see [Ezt13, Section 2.8]).

4. HODGE THEORY

In this section, we outline properties of Hermitian manifolds and important results, building towards the decomposition theorem. For a full treatment, see Chapters 6 and 7 of [GH14].

4.1. Laplace operators. Harmonic forms also satisfy another first order differential equation: the formal adjoint d^* of the operator d . We introduce the Laplacian operator which is defined in terms of these two operators before defining harmonic forms in Section 4.2.

Definition 4.1. For a complex vector space V , a *Hermitian form* is a complex-valued bilinear form $h: V \times V \rightarrow \mathbb{C}$ which is $\mathbb{C}$ -linear on the left and $\mathbb{C}$ -antilinear on the right, and satisfies $h(w, z) = \overline{h(z, w)}$. A *Hermitian metric* on a complex vector bundle $E \rightarrow X$ of a differentiable manifold X is a global section h of the smooth vector bundle $E^* \otimes E^*$ that on the fibre over any point p is a Hermitian form on the space E . If $E = T_{\mathbb{C}}X$ for a complex manifold X , then X is a (*complex*) *Hermitian manifold*. The real part $g = \frac{1}{2}(h + \bar{h})$ is a Riemannian metric on X and we call $\omega := -\frac{1}{2}\Im h$ the *associated* $(1-1)$ -form. If ω is a closed form, then we say that ω is a *symplectic form*, (X, h) is a *Kähler manifold*, and h is a *Kähler metric* on X .

Definition 4.2. Let E and F be complex vector bundles of a differentiable manifold X with volume form dv and equipped with Hermitian metrics $\langle \cdot, \cdot \rangle_E$ and $\langle \cdot, \cdot \rangle_F$ respectively. Let $P: C^\infty(E) \rightarrow C^\infty(F)$ and $Q: C^\infty(F) \rightarrow C^\infty(E)$ be $\mathbb{C}$ -linear derivations. Then Q is called the *formal adjoint* of P if

$$\int_X \langle P\alpha, \beta \rangle_F dv = \int_X \langle \alpha, Q\beta \rangle_E dv,$$

for all compactly supported smooth sections $\alpha \in C_c^\infty(E)$ and $\beta \in C_c^\infty(F)$.

For the proof of the following two lemmas, see [Mor07, Section 14.2].

Lemma 4.3. *On a Hermitian manifold (X, h) , there exists a Hermitian metric $H(\cdot, \cdot)$ on the complex vector bundle $\Omega_{\mathbb{C}}^k$ induced by the metric h and with local orthonormal basis $\{e_{i_1} \wedge \cdots \wedge e_{i_k} : 1 \leq i_1 < \cdots < i_k \leq n\}$, where $\{e_1, \dots, e_n\}$ is a local orthonormal frame for $\Omega_{\mathbb{C}}^1$.*

Lemma 4.4. *The exterior derivative $d: \Omega_{\mathbb{C}}^k(X) \rightarrow \Omega_{\mathbb{C}}^{k+1}(X)$ has a formal adjoint $\delta: \Omega_{\mathbb{C}}^{k+1}(X) \rightarrow \Omega_{\mathbb{C}}^k(X)$ with respect to the Hermitian metrics (both denoted $H(\cdot, \cdot)$) $\Omega_{\mathbb{C}}^k(X)$ and $\Omega_{\mathbb{C}}^{k+1}(X)$.*

Definition 4.5. The Laplace operator $\Delta: \Omega_{\mathbb{C}}^k(X) \rightarrow \Omega_{\mathbb{C}}^k(X)$ is defined by $\Delta := d\delta + \delta d$.

From the definition of formal adjoint, we see that if D has a formal adjoint D^* and C has a formal adjoint C^* , then $C \circ D$ has a formal adjoint $D^* \circ C^*$. Thus, $d\delta$ has a formal adjoint δd , so Δ is formally self-adjoint.

Lemma 4.6. *If X is a complex manifold equipped with a Hermitian metric, then δ decomposes into $\delta = \partial^* + \bar{\partial}^*$.*

Proof. See [Voi02, Section 5.1.3] □

Definition 4.7. For a complex manifold X equipped with a Hermitian metric, define Laplacians associated to the operators ∂ and $\bar{\partial}$ respectively as

$$\Delta_{\partial} := \partial\partial^* + \partial^*\partial \quad \text{and} \quad \Delta_{\bar{\partial}} := \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}.$$

Generally, the three Laplacian operators are distinct. The first Laplacian Δ only depends on the de Rham differential, whereas Δ_{∂} and $\Delta_{\bar{\partial}}$ depend on the complex structure of the manifold. On a Kähler manifold, there are many ways in which the real and complex structures are compatible, so it begs the question whether there is a nice relationship between the three in the Kähler case.

Theorem 4.8. *On a Kähler manifold, the Laplacian operators satisfy $\Delta = 2\Delta_{\partial} = 2\Delta_{\bar{\partial}}$.*

Proof. See [Voi02, Theorem 6.7]. □

4.2. Harmonic decompositions. For any differentiable manifold X , its de Rham cohomology groups are topological invariants. If in addition X is compact and has a Hermitian structure, we can exhibit particular representatives, called harmonic forms, for the de Rham cohomology classes. These harmonic forms play an important role in establishing properties of the Hodge decomposition.

Definition 4.9. Let X be a compact complex manifold. The *harmonic k -forms* are defined to be

$$\mathcal{H}^k(X; \mathbb{C}) := \{\omega \in \Omega_{\mathbb{C}}^k(X) : \Delta\omega = 0\}.$$

If furthermore (X, h) is a complex Hermitian manifold, then the *$\bar{\partial}$ -harmonic p, q -forms* are defined to be

$$\mathcal{H}^{p,q}(X) := \{\omega \in \Omega^{p,q}(X) : \Delta^{\bar{\partial}}\omega = 0\}.$$

Lemma 4.10. *Let X be a compact Hermitian manifold. A k -form is harmonic if and only if it is d -closed and δ -closed.*

Proof. If ω is both d -closed and δ -closed, then $\Delta\omega = (d\delta + \delta d)(\omega) = 0$. Conversely, since X is compact, and d and δ are formally adjoint, then for all k -forms η , $\int_X H(\delta\omega, \eta)dv = 0$. Letting $\eta = \omega$, we see that

$$0 = \int_X H(\Delta\omega, \omega)dv = \int_X H(d\delta\omega + \delta d\omega, \omega)dv = \int_X |\delta\omega|^2 + |d\omega|^2 dv.$$

Therefore, $d\omega = 0$ and $\delta\omega = 0$. $\square$

An analogous proof gives the following lemma:

Lemma 4.11. *Let X be a compact Hermitian manifold. A (p, q) -form is $\bar{\partial}$ -harmonic if and only if $\bar{\partial}\omega = 0$ and $\bar{\partial}^*\omega = 0$.*

Now we state the main theorems of Hodge theory, namely the decomposition theorems. The proofs are involved, and we refer the reader to [GH14, Chapter 6].

Theorem 4.12. *Let X be a compact Hermitian manifold. Then the space of (p, q) -forms decomposes as the direct sum*

$$\Omega_{\mathbb{C}}^k(X) = \mathcal{H}^k(X; \mathbb{C}) \oplus \delta\Omega_{\mathbb{C}}^{k+1}(X) \oplus d\Omega_{\mathbb{C}}^{k-1}(X).$$

Theorem 4.13. *Let X be a compact Hermitian manifold. The space of (p, q) -forms decomposes as the direct sum*

$$\Omega^{p,q}(X) = \mathcal{H}^{p,q}(X) \oplus \bar{\partial}^*\Omega^{p,q+1}(X) \oplus \bar{\partial}\Omega^{p,q-1}(X).$$

These decompositions yield tools for calculating de Rham and Dolbeault cohomologies. Since harmonic k -forms (resp. (p, q) -forms) are closed (resp. $\bar{\partial}$ -closed), there exists a natural map $\mathcal{H}^k(X; \mathbb{C}) \rightarrow H_{DR}^k(X; \mathbb{C})$ (resp. $\mathcal{H}^{p,q}(X) \rightarrow H^{p,q}(X)$) given by $\omega \mapsto [\omega]$.

Theorem 4.14 (Hodge and Dolbeault isomorphisms). *Let X be a compact Hermitian manifold. The natural maps $\mathcal{H}^k(X; \mathbb{C}) \rightarrow H_{DR}^k(X; \mathbb{C})$ and $\mathcal{H}^{p,q}(X) \rightarrow H^{p,q}(X)$ are isomorphisms.*

The Hodge decomposition theorem of Theorem 1.1 yields an explicit relationship between the complexified de Rham cohomology groups and the Dolbeault cohomology groups in the case when X is compact and Kähler.

Corollary 4.15 (Hodge symmetry). *$\overline{H^{p,q}(X)} = H^{q,p}(X)$, where complex conjugation acts naturally on $H^{p+q}(X, \mathbb{C}) = H^{p+q}(X, \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$.*

Corollary 4.16. *The odd Betti numbers $b_{2k+1} = \dim_{\mathbb{C}} H^{2k+1}(X, \mathbb{C})$ of a compact Kähler manifold are even.*

By Theorem 4.14 and Example 3.4, we conclude this section with the following result:

Lemma 4.17. *$H^{p,q}(X)$ is canonically isomorphic to $H^q(X, \Omega_{hol}^p)$.*

5. DEGENERATION OF THE FRÖLICHER SPECTRAL SEQUENCE AND HODGE DECOMPOSITION

Suppose X is an n -dimensional complex manifold and let $\underline{\mathbb{C}}$ be the locally constant sheaf of stalk $\mathbb{C}$ over X . The sheaves of holomorphic p -forms Ω_{hol}^p on X forms a complex

$$0 \rightarrow \Omega_{hol}^0 \xrightarrow{\partial} \Omega_{hol}^1 \xrightarrow{\partial} \Omega_{hol}^2 \xrightarrow{\partial} \dots \xrightarrow{\partial} \Omega_{hol}^n \rightarrow 0,$$

called the *holomorphic de Rham complex*, with $\partial: \Omega_{hol}^q \rightarrow \Omega_{hol}^{q+1}$ being the exterior differentiation operator. This complex forms a resolution of $\underline{\mathbb{C}}$ by the holomorphic Poincaré lemma [Voi02, Proposition 2.31], so it follows then that $H^k(X, \underline{\mathbb{C}}) = \mathbb{H}^k(X, \Omega_{hol}^\bullet)$ as we can view $\underline{\mathbb{C}}$ as a complex supported in degree zero. We can equip $(\Omega_{hol}^\bullet, \partial)$ with the naïve filtration

$$F^p\Omega_{hol}^\bullet := [0 \rightarrow \Omega_{hol}^p \rightarrow \Omega_{hol}^{p+1} \rightarrow \dots \rightarrow \Omega_{hol}^n \rightarrow 0],$$

which induces a spectral sequence called the *Frölicher spectral sequence*, or the *Hodge to de Rham spectral sequence*, given by

$$E_i^{p,q} \Rightarrow \mathbb{H}^{p+q}(X, \Omega_{hol}^\bullet) = H^{p+q}(X, \mathbb{C}).$$

This allows us to give the following definition:

Definition 5.1. The *Hodge filtration* on the cohomology of a complex manifold is

$$F^p H^k(X, \mathbb{C}) := \text{im}(\mathbb{H}^k(X, F^p \Omega_{hol}^\bullet) \rightarrow \mathbb{H}^k(X, \Omega_{hol}^\bullet)).$$

Concretely, we can take the filtration of the double complex associated to the Dolbeault resolution $(\mathcal{A}^{q,\cdot}, \bar{\partial})$ of Ω_X^q for $q \geq p$ given by

$$F^p \mathcal{A}^k = \bigoplus_{r \geq p} \mathcal{A}^{r, k-r}.$$

Taking the cohomology of the complex of its global sections $(F^p \mathcal{A}^\bullet(X), d)$ yields the hypercohomology of $F^p \Omega_X^\bullet$. The first page of the Frölicher spectral sequence is simply the cohomology of the differential $\bar{\partial}$, so $E_1^{p,q} = H^q(\mathcal{A}(X)^{p,\bullet}, \bar{\partial})$. We know this to be the normal Dolbeault cohomology, and by 4.17, we have $H^q(\mathcal{A}^{p,\bullet}(X), \bar{\partial}) = H^q(X, \Omega_X^p)$. The differential $d_1: E_1^{p,1} \rightarrow E_1^{p+1,q}$ is induced by ∂ , and thus is simply the map $\partial: H^1(X, \Omega_X^p) \rightarrow H^1(X, \Omega_X^{p+1})$. Therefore, the Hodge filtration on cohomology is also given by

$$F^p H^k(X, \mathbb{C}) = \bigoplus_{r \geq p} H^{r, k-r}(X),$$

the filtration on the Hodge decomposition. We also have a natural isomorphism

$$\text{Gr}_F^p H^{p+q}(X, \mathbb{C}) = \frac{\bigoplus_{r \geq p} H^{r, k-r}(X)}{\bigoplus_{r \geq p+1} H^{r, k-r}(X)} = H^{p,q}(X) \cong H^q(X, \Omega_X^p).$$

We can now deduce Theorem 1.2.

Proof of Theorem 1.2. For sufficiently large r , we have that

$$\text{Gr}_F^p H^k(X, \mathbb{C}) = E_\infty^{p,q} = E_r^{p,q}.$$

Now $\dim E_i^{p,q} \leq \dim E_{i-1}^{p,q}$ as each $E_i^{p,q}$ is computed by taking the cohomology $\ker d_i / \text{im } d_i$ in bidegree (p, q) , with equality for every p, q if and only if $d_i = 0$. Thus,

$$\dim E_\infty^{p,q} = \dim \text{Gr}_F^p H^{p+q}(X, \mathbb{C}) = \dim H^q(X, \Omega_X^p) = \dim E_1^{p,q},$$

thus showing that $d_i = 0$ for all $i \geq 1$. Therefore, the spectral sequence degenerates at the E_1 page. $\square$

We now see how the degeneracy at E_1 of the Frölicher spectral sequence does not imply Theorem 1.1.

Example 5.2. For any compact complex surface X , we have the short exact sequence

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathcal{O}_X \longrightarrow \Omega_{hol}^1 \longrightarrow 0,$$

where $\mathcal{O}_X$ is the structure sheaf of X . We obtain the exact sequence

$$0 \longrightarrow H^0(X, \Omega_{hol}^1) \longrightarrow H^1(X, \mathbb{C}) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow 0$$

from cohomology (see [BHPVdV15, Lemma 2.6]). Consider the Hopf surface $X = (\mathbb{C}^2 \setminus \{0\})/\mathbb{Z}$, where $k \in \mathbb{Z}$ acts by scalar multiplication by λ^k for some $\lambda \in \mathbb{C}^*$ and $|\lambda| < 1$, which is not Kähler. We know that X is diffeomorphic to $S^1 \times S^3$ [BHPVdV15, Theorem 18.1(i)], so it can be shown that $H^1(X, \mathbb{C}) = \mathbb{C}$. Furthermore, $H^0(X, \Omega_{hol}^1) = 0$ since if $\alpha \neq 0$ were a holomorphic 1-form, it is closed, and $\bar{\alpha}$ is not $\bar{\partial}$ -exact since there are no non-constant pluriharmonic functions on X . This implies that $H^1(X, \mathcal{O}_X) \neq 0$. On the other hand, there are no non-zero closed form of type $(0, 1)$, since if β is such a form, $\bar{\beta}$ is holomorphic. We therefore see that Theorem 1.1 is stronger than Theorem 1.2.

REFERENCES

- [Bei24] Yangda Bei. The algebraic hodge to de rham spectral sequence. 2024.
- [BHPVdV15] Wolf Barth, Klaus Hulek, Chris Peters, and Antonius Van de Ven. *Compact complex surfaces*, volume 4. Springer, 2015.
- [EZT13] Fouad El Zein and Loring W Tu. From sheaf cohomology to the algebraic de rham theorem. *Hodge theory*, 49:70–122, 2013.
- [GH14] Phillip Griffiths and Joseph Harris. *Principles of algebraic geometry*. John Wiley & Sons, 2014.
- [Mor07] Andrei Moroianu. *Lectures on Kähler geometry*, volume 69. Cambridge University Press, 2007.
- [Ser56] Jean-Pierre Serre. Géométrie algébrique et géométrie analytique. In *Annales de l’institut Fourier*, volume 6, pages 1–42, 1956.
- [Voi02] Claire Voisin. *Hodge Theory and Complex Algebraic Geometry I: Volume 1*, volume 76. Cambridge University Press, 2002.